

clinical trials at many different locations, the standard therapy was administered to patients with comparable hypertension (as measured by the New York Heart Association (NYHA) Classification). The lab then classified the responses to therapy for this large patient group into one of four response categories. Table 10.1 lists the categories and percentages of patients treated on the standard preparation who have been classified in each category.

TABLE 10.1
Results of clinical trials using
the standard preparation

Category	Percentage
Marked decrease in blood pressure	50
Moderate decrease in blood pressure	25
Slight decrease in blood pressure	10
Stationary or slight increase in blood pressure	15

The lab then conducted a clinical trial with a random sample of 200 patients with high blood pressure. All patients were required to be listed according to the same hypertensive categories of the NYHA Classification as those studied under the standard preparation. Use the sample data in Table 10.2 to test the hypothesis that the cell probabilities associated with the test preparation are identical to those for the standard. Use $\alpha = .05$.

TABLE 10.2
Sample data for
example

Category	Observed Cell Counts
1	120
2	60
3	10
4	10

Solution This experiment possesses the characteristics of a multinomial experiment, with $n = 200$ and $k = 4$ outcomes.

- Outcome 1: A person's blood pressure will decrease markedly after treatment with the test drug.
- Outcome 2: A person's blood pressure will decrease moderately after treatment with the test drug.
- Outcome 3: A person's blood pressure will decrease slightly after treatment with the test drug.
- Outcome 4: A person's blood pressure will remain stationary or increase slightly after treatment with the test drug.

The null and alternative hypotheses are then

$$H_0: \pi_1 = .50, \pi_2 = .25, \pi_3 = .10, \pi_4 = .15$$

and

$$H_a: \text{At least one of the cell probabilities is different from the hypothesized value.}$$

Before computing the test statistic, we must determine the expected cell numbers. These data are given in Table 10.3.

TABLE 10.3

Observed and expected cell numbers for example

Category	Observed Cell Number, n_i	Expected Cell Number, E_i
1	120	$200(.50) = 100$
2	60	$200(.25) = 50$
3	10	$200(.10) = 20$
4	10	$200(.15) = 30$

Because all the expected cell numbers are relatively large, we may calculate the chi-square statistic and compare it to a tabulated value of the chi-square distribution.

$$\begin{aligned}
 \chi^2 &= \sum_i \left[\frac{(n_i - E_i)^2}{E_i} \right] \\
 &= \frac{(120 - 100)^2}{100} + \frac{(60 - 50)^2}{50} + \frac{(10 - 20)^2}{20} + \frac{(10 - 30)^2}{30} \\
 &= 4 + 2 + 5 + 13.33 = 24.33
 \end{aligned}$$

For the probability of a Type I error set at $\alpha = .05$, we look up the value of the chi-square statistic for $\alpha = .05$ and $df = k - 1 = 3$. The critical value from Table 7 in the Appendix is 7.815.

R.R.: Reject H_0 if $\chi^2 > 7.815$.

Conclusion: The computed value of χ^2 is greater than 7.815, so we reject the null hypothesis and conclude that at least one of the cell probabilities differs from that specified under H_0 . Practically, it appears that a much higher proportion of patients treated with the test preparation falls into the moderate and marked improvement categories. The p -value for this test is $p < .001$. (See Table 7 in the Appendix.)

The assumptions needed for running a chi-square goodness-of-fit test are those associated with a multinomial experiment, of which the key ones are independence of the trials and constant cell probabilities. Independence of the trials would be violated if, for example, several patients from the same family were included in the sample because hypertension has a strong hereditary component. The assumption of constant cell probabilities would be violated if the study were conducted over a period of time during which the standards of medical practice shifted, allowing for other “standard” therapies.

The test statistic for the chi-square goodness-of-fit test is the sum of k terms, which is the reason the degrees of freedom depend on k , the number of categories, rather than on n , the total sample size. However, there are only $k - 1$ degrees of freedom, rather than k , because the sum of the $n_i - E_i$ terms must be equal to $n - n = 0$; $k - 1$ of the observed minus expected differences are free to vary, but the last one (k th) is determined by the condition that the sum of the $n_i - E_i$ equals zero.

This goodness-of-fit test has been used extensively over the years to test various scientific theories. Unlike previous statistical tests, however, the hypothesis of interest is the null hypothesis, not the research (or alternative) hypothesis. Unfortunately, the logic behind running a statistical test does not hold. In the